

TRIGONOMETRY

Sine Rule & Cosine Rule — Grade 6 Lesson

Learning with Fun & Curiosity! ■

1. What is Trigonometry?

Trigonometry helps us find **unknown sides or angles** in triangles. Think of it like being a detective for shapes! When we know *some* parts of a triangle, trigonometry helps us discover the missing pieces.

Every triangle has **3 sides** (labelled a , b , c) and **3 angles** (labelled A , B , C). The side opposite angle A is called side a , and so on.

2. The Sine Rule

$$a / \sin A = b / \sin B = c / \sin C$$

When to use the Sine Rule:

Situation	What you know
AAS / ASA	Two angles + one side
SSA	Two sides + angle opposite one of them

Worked Example — Sine Rule:

Problem: Find side b if $a = 7$ cm, $\angle A = 50^\circ$, $\angle B = 60^\circ$

Step 1	Write the Sine Rule: $b / \sin B = a / \sin A$
Step 2	Substitute values: $b / \sin 60^\circ = 7 / \sin 50^\circ$
Step 3	Solve for b : $b = 7 \times \sin 60^\circ / \sin 50^\circ$
Step 4	Calculate: $b \approx 7 \times 0.866 / 0.766 \approx 7.91$ cm

3. The Cosine Rule

$$c^2 = a^2 + b^2 - 2ab \cos C$$

To find an angle when all three sides are known, rearrange to:

$$\cos C = (a^2 + b^2 - c^2) / (2ab)$$

When to use the Cosine Rule:

Situation	What you know
SAS	Two sides + the included angle between them
SSS	All three sides (to find any angle)

Worked Example — Cosine Rule:

Problem: Find side c if $a = 5$ cm, $b = 6$ cm, $\angle C = 60^\circ$

Step 1	Write the Cosine Rule: $c^2 = a^2 + b^2 - 2ab \cos C$
Step 2	Substitute values: $c^2 = 5^2 + 6^2 - 2(5)(6) \cos 60^\circ$
Step 3	Calculate parts: $c^2 = 25 + 36 - 60 \times 0.5 = 31$
Step 4	Take square root: $c = \sqrt{31} \approx 5.57$ cm

4. Sine Rule vs Cosine Rule — Quick Guide

Feature	Sine Rule	Cosine Rule
Use when	AAS, ASA, SSA	SAS, SSS
Formula type	Ratios	Squares & cosine
Key formula	$a/\sin A = b/\sin B = c/\sin C$	$c^2 = a^2 + b^2 - 2ab \cos C$
Good for	Finding sides or angles with angle-side pairs	Included angles or all sides known

5. Practice Problems

Part A — Sine Rule

Q1	Find side b if $a = 8$ cm, $\angle A = 45^\circ$, $\angle B = 70^\circ$	Answer: _____
Q2	Find $\angle B$ if $a = 10$ cm, $b = 12$ cm, $\angle A = 40^\circ$	Answer: _____
Q3	Find side c if $\angle A = 55^\circ$, $\angle C = 65^\circ$, $a = 9$ cm	Answer: _____

Part B — Cosine Rule

Q1	Find side c if $a = 8$ cm, $b = 5$ cm, $\angle C = 45^\circ$	Answer: _____
Q2	Find $\angle A$ if $a = 7$ cm, $b = 9$ cm, $c = 6$ cm	Answer: _____
Q3	Find side b if $a = 10$ cm, $c = 12$ cm, $\angle B = 70^\circ$	Answer: _____

6. Real-Life Applications

■ Architecture	Engineers use sine and cosine rules to calculate roof angles, beam lengths, and building heights with precision.
■ Navigation	Sailors and pilots apply these rules to find distances between locations and determine the correct direction of travel.
■ Sports	Coaches use trigonometry to measure angles for optimal throws, kicks, and jumps to improve athletic performance.

Key Takeaways

- The **Sine Rule** uses ratios: $a/\sin A = b/\sin B = c/\sin C$ (use for AAS, ASA, SSA)
- The **Cosine Rule** uses squares: $c^2 = a^2 + b^2 - 2ab \cos C$ (use for SAS, SSS)
- **Both rules** work for *any* triangle — not just right-angled ones!
- Always check **which information you have** before choosing the rule.

Keep practicing and stay curious! Trigonometry is your superpower for solving triangle mysteries. ■